Customer Shopping Behavior Analysis

End-to-end data analytics project transforming retail transaction data into actionable business insights

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Project Overview

01

Data Cleaning & EDA

Python (Jupyter Notebook) for preprocessing, exploration, and feature engineering

02

SQL Business Analysis

PostgreSQL for answering key questions on spending, preferences, and loyalty

03

Interactive Dashboard

Power BI for visualizing insights and supporting data-driven decisions

Dataset: **3,900 rows × 18 columns** covering demographics, transactions, preferences, and behavior

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Business Problem

Key Question

How can customer shopping data identify trends, improve engagement, and optimize marketing and product strategies?

Objective

Understand customer demographics, purchasing habits, category performance, subscription behavior, and promotional impact to drive better decision-making.

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SQL Analysis Highlights

Gender Revenue

Male customers generated **$157,890** vs. female **$75,191**

Age Groups

60+ and 50s age groups contributed highest revenue at ~$46K each

Subscription Impact

Non-subscribers: 2,847 customers, $170,436 revenue.
Subscribers: 1,053 customers, $62,645 revenue

Discount Strategy

Hats (50%), Sneakers (49.66%), Coats (49.07%) most frequently purchased with discounts

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Product Performance Insights

Top Products by Rating

Gloves (3.86), Sandals (3.84), Boots (3.82), Hats (3.80), Skirts (3.79)

Top Products by Sales

Blouses, Pants, Shirts (Clothing); Sandals, Shoes, Sneakers (Footwear); Jewelry, Sunglasses, Belts (Accessories)

Shipping Impact

Express shipping ($60.48 avg) vs. Standard ($58.46 avg) shows minimal difference in purchase value

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customer behavior dashboard • Last saved: Today at 6:04 PM

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File | Home | Insert | Modeling | View | Optimize | Help

Share

Paste | Cut | Copy | Format painter — Clipboard

Get data | Excel workbook | OneLake catalog | SQL Server | Enter data | Dataverse | Recent sources — Data

Transform data | Refresh — Queries

New visual | Text box | More visuals — Insert

New visual calculation | New measure | Quick measure — Calculations

Sensitivity — Sensitivity

Publish — Share

Prep data for AI | Copilot — Copilot

Customer Shopping Behavior Dashboard

Subscription Status

No | Yes

Gender

Female | Male

Category

Accessories

Clothing

Footwear

Shipping Type

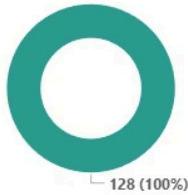
- 2-Day Shipping
- Express
- Free Shipping
- Next Day Air
- Standard
- Store Pickup

128
Number Of Customers

$57.30
Average Purchase Amount

3.63
Average Review Rating

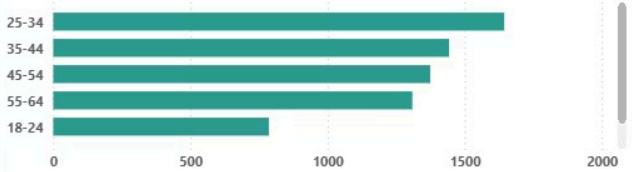
% of customers by Subscription Status



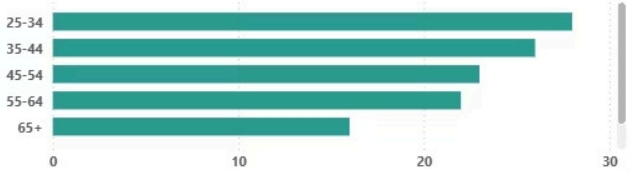
Sales by Category

Revenue by Category

Sum of Purchase Amount (USD) by Age Group



Sales by Age Group



Filters | Visualizations | Data

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Mostly sunny

Search

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27-03-2026

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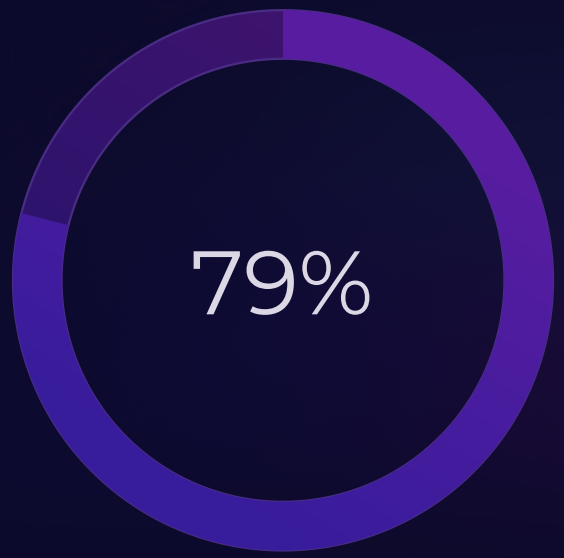
Customer Segmentation

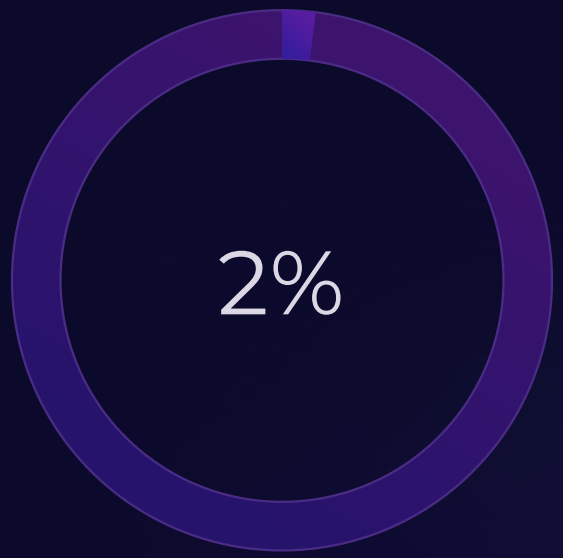


Loyal Customers

3,116 customers with >10 previous purchases

Returning

701 customers with 2-10 previous purchases



New

83 customers with only 1 previous purchase

Repeat buyers (previous purchases >5) show higher subscription rates: 958 subscribed vs. 2,518 non-subscribed

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Key Business Questions Answered

- Which gender contributes more to total revenue?
- Which age groups generate the highest purchase value?
- What are the top-performing product categories by revenue?
- Which products receive the highest average review ratings?
- Do subscribers spend more than non-subscribers?
- Are repeat buyers more likely to subscribe?
- Which product categories rely most heavily on discounts?
- Which payment methods are most preferred by customers?
- Which season generates the highest sales?
- How do shipping types impact average purchase value?

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Business Recommendations

1

Strengthen Loyalty Programs

Target repeat buyers with exclusive offers, rewards, and membership benefits

2

Increase Subscription Adoption

Promote plans with early access, exclusive discounts, free shipping, and loyalty points

3

Optimize Discount Strategy

Use strategically for low-conversion categories and seasonal campaigns; avoid over-discounting top performers

4

Focus on High-Value Segments

Prioritize marketing to high-spending age groups, repeat customers, and subscribers

5

Improve Product Positioning

Feature top-rated products prominently in campaigns and prioritize stock planning

6

Leverage Seasonal Trends

Align promotional calendars, inventory allocation, and product bundling with seasonal patterns

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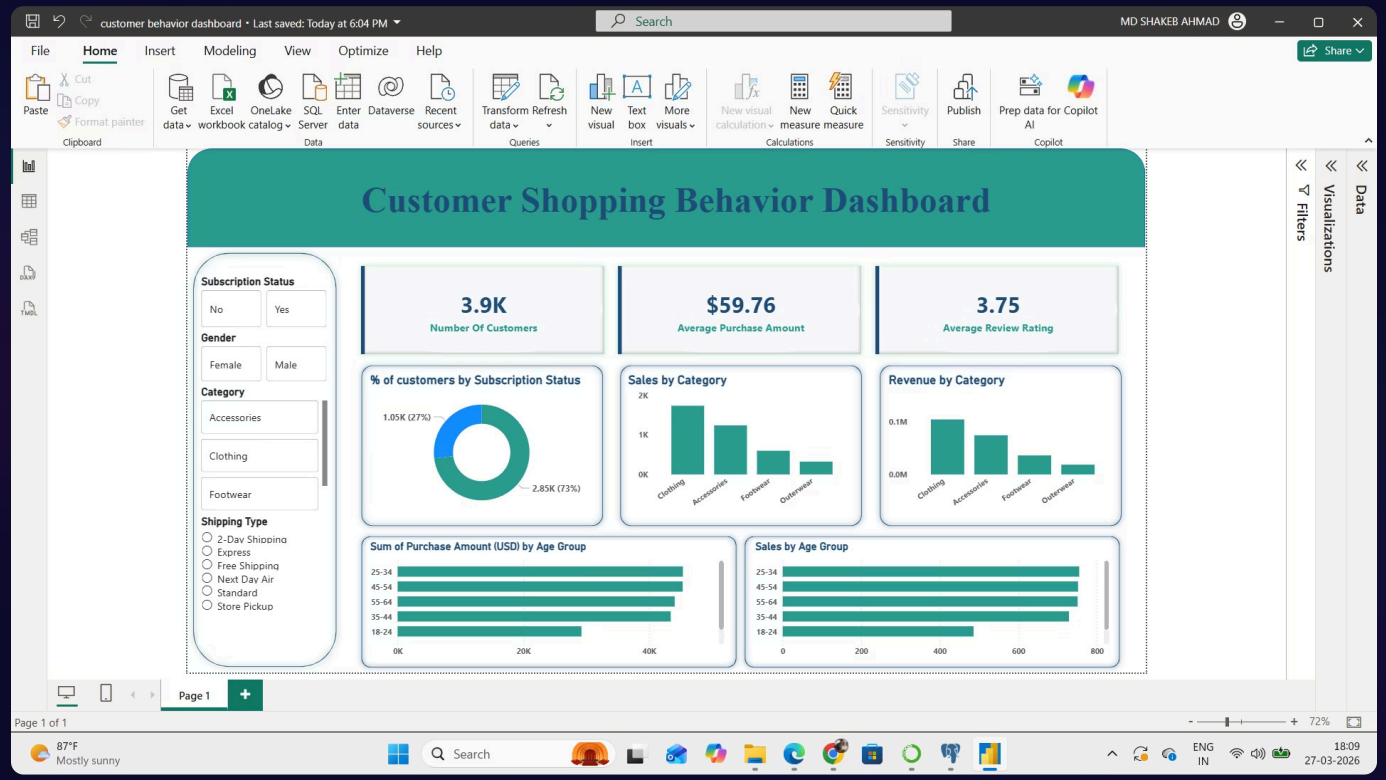
Dashboard & Visualization

Interactive Features

- Customer demographics by age, gender, location
- Purchase behavior trends and patterns
- Category performance metrics
- Subscription and loyalty insights
- Discount and promotion impact analysis
- Seasonal and payment method patterns



Power BI dashboard enables stakeholders to interactively explore key metrics and make informed business decisions

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Project Impact

Complete Workflow

End-to-end data analytics from cleaning to visualization

Actionable Insights

Transformed raw data into strategic business recommendations

Portfolio Showcase

Real-world application of Python, SQL, and Power BI skills

This project demonstrates how customer transaction data drives improved targeting, increased repeat purchases, optimized promotions, and stronger loyalty programs.

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